

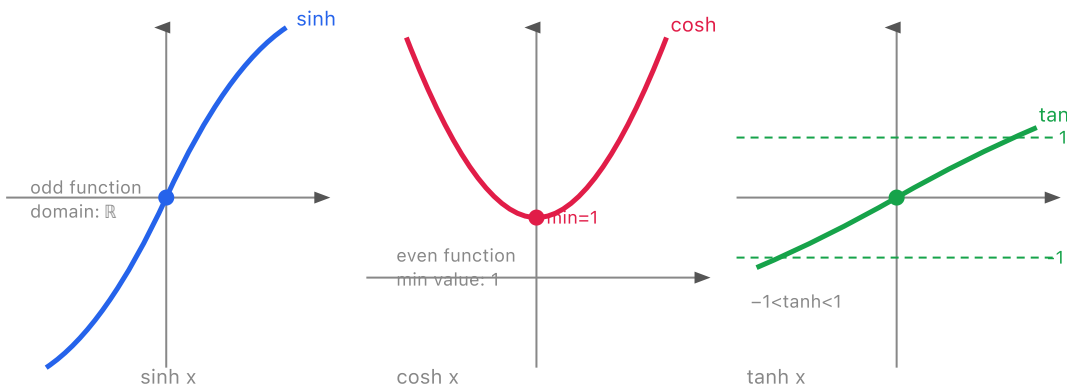
1.8: Hyperbolic Functions

1. Definitions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

The reciprocal functions: $\operatorname{cosech} x = \frac{1}{\sinh x}$, $\operatorname{sech} x = \frac{1}{\cosh x}$, $\operatorname{coth} x = \frac{1}{\tanh x}$

2. Graphs of Hyperbolic Functions



Key Properties Summary

Function	Domain	Range	Parity	Key values
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	Odd	$\sinh 0 = 0$

$\cosh x$	$\mathbb{R}$	$[1, \infty)$	Even	$\cosh 0 = 1$, min at $x = 0$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$	Odd	$\tanh 0 = 0$; asymptotes $y = \pm 1$

3. Hyperbolic Identities

Fundamental identity: $\cosh^2 x - \sinh^2 x = 1$

Proof:
$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 =$$

$$\frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{4} = \frac{4}{4} = 1$$

Further Identities

Hyperbolic identity	Compare with trig identity
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$
$\sinh(2x) = 2 \sinh x \cosh x$	$\sin(2x) = 2 \sin x \cos x$
$\cosh(2x) = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x$	$\cos(2x) = \cos^2 x - \sin^2 x$
$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$	$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$	$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$1 + \tan^2 x = \sec^2 x$

Osborne's Rule

Osborne's rule: To convert a trigonometric identity to the corresponding hyperbolic identity: Replace $\cos$ with $\cosh$, replace $\sin$ with $\sinh$, **but replace every product $\sin^2$** (or any product of two sines) with $-\sinh^2$ (negate).

Note the sign change: $\cos^2 \theta + \sin^2 \theta = 1$ becomes $\cosh^2 x - \sinh^2 x = 1$ (the product $\sin \cdot \sin$ picks up a minus sign).

AQA Exam Tip: Osborne's rule is a memory aid, not a proof. AQA may ask you to "prove" an identity — use the exponential definitions directly rather than citing Osborne's rule.

4. Inverse Hyperbolic Functions

Logarithmic forms:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad x \in \mathbb{R}$$

$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1$$

$$\operatorname{arctanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1$$

Derivation of $\operatorname{arcsinh} x$: Let $y = \sinh^{-1} x$, so $\sinh y = x$, i.e. $\frac{e^y - e^{-y}}{2} = x$.

$e^{2y} - 2xe^y - 1 = 0$. Quadratic in e^y :

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$$

Since $e^y > 0$ for all y , take the positive root: $e^y = x + \sqrt{x^2 + 1}$

$$y = \ln(x + \sqrt{x^2 + 1}). \quad \checkmark$$

5. Derivatives

Derivatives of hyperbolic functions:

$$\frac{d}{dx}(\sinh x) = \cosh x, \quad \frac{d}{dx}(\cosh x) = \sinh x, \quad \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

Derivatives of inverse hyperbolic functions:

$$\frac{d}{dx}(\operatorname{arcsinh} x) = \frac{1}{\sqrt{x^2 + 1}}, \quad \frac{d}{dx}(\operatorname{arccosh} x) = \frac{1}{\sqrt{x^2 - 1}}, \quad \frac{d}{dx}(\operatorname{arctanh} x) = \frac{1}{1 - x^2}$$

AQA Exam Tip: Note $\frac{d}{dx}(\cosh x) = \sinh x$ (positive), unlike $\frac{d}{dx}(\cos x) = -\sin x$. This difference catches many students out.

6. Integration Using Hyperbolic Functions

Standard integrals (inverse hyperbolic):

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \operatorname{arcsinh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 + a^2}\right) + C$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \operatorname{arccosh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 - a^2}\right) + C, \quad x > a$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{a} \operatorname{arctanh}\left(\frac{x}{a}\right) + C = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + C, \quad |x| < a$$

Example: Find $\int \frac{1}{\sqrt{x^2 + 4}} dx$.

Using formula with $a = 2$: $\operatorname{arcsinh}\left(\frac{x}{2}\right) + C = \ln\left(x + \sqrt{x^2 + 4}\right) + C$

Example: Find $\int \sinh^2 x dx$.

Use identity: $\sinh^2 x = \frac{\cosh 2x - 1}{2}$

$$\int \frac{\cosh 2x - 1}{2} dx = \frac{\sinh 2x}{4} - \frac{x}{2} + C$$

Hyperbolic Substitutions

For integrals involving $\sqrt{x^2 + a^2}$, use the substitution $x = a \sinh t$ (since $a^2 \sinh^2 t + a^2 = a^2 \cosh^2 t$).

For integrals involving $\sqrt{x^2 - a^2}$, use $x = a \cosh t$.

Example: Find $\int \sqrt{x^2 + 1} dx$ using $x = \sinh t$.

$$\begin{aligned} dx &= \cosh t dt, \sqrt{x^2 + 1} = \sqrt{\sinh^2 t + 1} = \cosh t \\ \int \cosh^2 t dt &= \int \frac{\cosh 2t + 1}{2} dt = \frac{\sinh 2t}{4} + \frac{t}{2} + C \\ &= \frac{\sinh t \cosh t}{2} + \frac{t}{2} + C = \frac{x\sqrt{x^2 + 1}}{2} + \frac{1}{2} \operatorname{arcsinh} x + C \end{aligned}$$

7. Solving Hyperbolic Equations

Example: Solve $5 \cosh x - 3 \sinh x = 7$.

$$\begin{aligned} \text{Substitute definitions: } 5 \cdot \frac{e^x + e^{-x}}{2} - 3 \cdot \frac{e^x - e^{-x}}{2} &= 7 \\ \frac{5e^x + 5e^{-x} - 3e^x + 3e^{-x}}{2} &= 7 \\ 2e^x + 8e^{-x} &= 14 \\ e^{2x} - 7e^x + 4 &= 0 \text{ (multiply by } e^x/2) \\ e^x &= \frac{7 \pm \sqrt{49 - 16}}{2} = \frac{7 \pm \sqrt{33}}{2} \\ x &= \ln\left(\frac{7 + \sqrt{33}}{2}\right) \text{ or } x = \ln\left(\frac{7 - \sqrt{33}}{2}\right) \end{aligned}$$

8. Summary of Key Results

Result	Formula
Fundamental identity	$\cosh^2 x - \sinh^2 x = 1$
$\operatorname{arcsinh} x$	$\ln(x + \sqrt{x^2 + 1})$
$\operatorname{arccosh} x$	$\ln(x + \sqrt{x^2 - 1}), x \geq 1$
$\operatorname{arctanh} x$	$\frac{1}{2} \ln \frac{1+x}{1-x}, x < 1$
$\frac{d}{dx} \sinh x$	$\cosh x$
$\frac{d}{dx} \cosh x$	$\sinh x$
$\frac{d}{dx} \tanh x$	$\operatorname{sech}^2 x$
$\int 1/\sqrt{x^2 + a^2} dx$	$\operatorname{arcsinh}(x/a) + C$
$\int 1/\sqrt{x^2 - a^2} dx$	$\operatorname{arccosh}(x/a) + C$

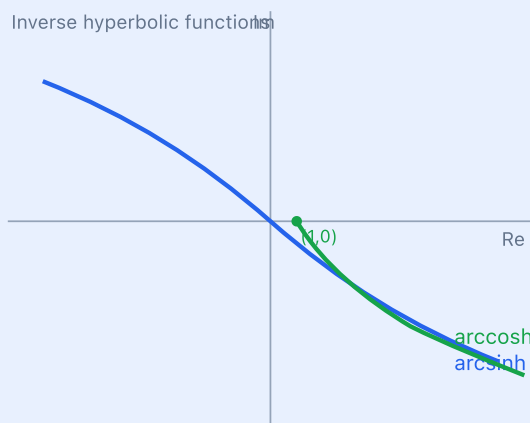
9. Practice Checklist

- Can you define $\sinh$, $\cosh$, $\tanh$ in terms of exponentials?
- Can you prove the identity $\cosh^2 x - \sinh^2 x = 1$ from the definitions?
- Can you state and use the double-angle hyperbolic identities?
- Can you derive the logarithmic forms of the inverse functions?
- Can you differentiate and integrate hyperbolic and inverse hyperbolic functions?
- Can you solve equations like $a \cosh x + b \sinh x = c$ by substituting exponential definitions?

Inverse Hyperbolic Functions — Graphs and Domains

Function	Domain	Range	Logarithmic form
----------	--------	-------	------------------

$\operatorname{arcsinh} x$	$\mathbb{R}$	$\mathbb{R}$	$\ln(x + \sqrt{x^2 + 1})$
$\operatorname{arccosh} x$	$x \geq 1$	$y \geq 0$	$\ln(x + \sqrt{x^2 - 1})$
$\operatorname{arctanh} x$	$ x < 1$	$\mathbb{R}$	$\frac{1}{2} \ln \frac{1+x}{1-x}$



Derivatives of inverse hyperbolics:

$$\frac{d}{dx} \operatorname{arcsinh} x = \frac{1}{\sqrt{x^2 + 1}}, \quad \frac{d}{dx} \operatorname{arccosh} x = \frac{1}{\sqrt{x^2 - 1}}, \quad \frac{d}{dx} \operatorname{arctanh} x = \frac{1}{1 - x^2}$$

Integration Using Hyperbolic Functions

Hyperbolic substitutions and direct standard forms:

Example 1: $\int \frac{1}{\sqrt{x^2 + 4}} dx$. Let $x = 2 \sinh t$, $dx = 2 \cosh t dt$, $\sqrt{x^2 + 4} = 2 \cosh t$:

$$\int \frac{2 \cosh t}{2 \cosh t} dt = t + C = \operatorname{arcsinh}(x/2) + C = \ln \left(\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1} \right) + C$$

Example 2: $\int \frac{1}{4 - x^2} dx$ for $|x| < 2$. Let $x = 2 \tanh t$, $dx = 2 \operatorname{sech}^2 t dt$, $4 - x^2 = 4 \operatorname{sech}^2 t$:

$$\int \frac{2\operatorname{sech}^2 t}{4\operatorname{sech}^2 t} dt = \frac{t}{2} + C = \frac{1}{2} \operatorname{arctanh}(x/2) + C = \frac{1}{4} \ln \frac{2+x}{2-x} + C$$

Example 3: $\int_0^1 \sqrt{1+x^2} dx$. Let $x = \sinh t$: $dx = \cosh t dt$, limits 0 to $\operatorname{arcsinh}(1) = \ln(1 + \sqrt{2})$:

$$\int_0^{\ln(1+\sqrt{2})} \cosh^2 t dt = \left[\frac{\sinh 2t}{4} + \frac{t}{2} \right]_0^{\ln(1+\sqrt{2})} = \frac{\sqrt{2}}{2} + \frac{\ln(1 + \sqrt{2})}{2}$$

The Catenary — Application of cosh

A uniform flexible chain hanging under gravity takes the shape of a **catenary**:

$$y = a \cosh\left(\frac{x}{a}\right)$$

where a is the height of the lowest point above the x -axis direction. Properties:

- Minimum at $(0, a)$ with $y'(0) = 0$.
- $y'' = \frac{1}{a} \cosh(x/a) = \frac{y}{a^2}$ — curvature is proportional to y .
- Arc length from $-L$ to L : $2a \sinh(L/a)$.
- The Gateway Arch in St Louis is (approximately) an inverted catenary.

Surface area of a catenary: When $y = \cosh x$ is rotated about the x -axis from $-a$ to a :

$$S = 2\pi \int_{-a}^a \cosh x \sqrt{1 + \sinh^2 x} dx = 2\pi \int_{-a}^a \cosh^2 x dx = \pi [\sinh(2a) + 2a]$$

(using the double-angle identity $\cosh^2 x = (1 + \cosh 2x)/2$).

AQA Exam Technique — Hyperbolic Functions

Key tips:

- **Identity proofs:** Start from exponential definitions. Don't try to prove by analogy with trig — show the working explicitly.
- **Osborne's rule:** Replace $\cos \rightarrow \cosh$, $\sin \rightarrow \sinh$, but replace $\sin^2 \rightarrow -\sinh^2$ (change sign when $\sin^2$ appears). This gives the correct hyperbolic identities.
- **Inverse forms:** For the logarithmic derivation, set $y = \sinh^{-1} x$ so $\sinh y = x$, write as exponential, solve the quadratic in e^y , discard the negative root.
- **Integration:** Recognise $\frac{1}{\sqrt{a^2 + x^2}} \rightarrow \operatorname{arcsinh}(x/a)$ and $\frac{1}{\sqrt{x^2 - a^2}} \rightarrow \operatorname{arccosh}(x/a)$ as standard forms.
- **Equations** $a \cosh x + b \sinh x = c$: Substitute exponential forms and rearrange to a quadratic in e^x . Discard solutions where $e^x \leq 0$.

Standard integral	Result
$\int \sinh x \, dx$	$\cosh x + C$
$\int \cosh x \, dx$	$\sinh x + C$
$\int \tanh x \, dx$	$\ln(\cosh x) + C$
$\int \frac{1}{\sqrt{x^2 + a^2}} \, dx$	$\operatorname{arcsinh}(x/a) + C$
$\int \frac{1}{\sqrt{x^2 - a^2}} \, dx$	$\operatorname{arccosh}(x/a) + C$
$\int \frac{1}{a^2 - x^2} \, dx$	$\frac{1}{a} \operatorname{arctanh}(x/a) + C = \frac{1}{2a} \ln \frac{a+x}{a-x} + C$

Hyperbolic Identities — Proofs

All hyperbolic identities follow from the definitions $\cosh x = \frac{e^x + e^{-x}}{2}$, $\sinh x = \frac{e^x - e^{-x}}{2}$.

Proof of $\cosh^2 x - \sinh^2 x = 1$:

$$\cosh^2 x = \frac{e^{2x} + 2 + e^{-2x}}{4}, \quad \sinh^2 x = \frac{e^{2x} - 2 + e^{-2x}}{4}.$$

$$\cosh^2 x - \sinh^2 x = \frac{4}{4} = 1. \quad \checkmark$$

Proof of $\sinh(A + B) = \sinh A \cosh B + \cosh A \sinh B$:

$$\sinh A \cosh B = \frac{e^A - e^{-A}}{2} \cdot \frac{e^B + e^{-B}}{2} = \frac{e^{A+B} + e^{A-B} - e^{-A+B} - e^{-(A+B)}}{4}$$

$$\cosh A \sinh B = \frac{e^{A+B} - e^{A-B} + e^{-A+B} - e^{-(A+B)}}{4}$$

$$\text{Adding: } \frac{2e^{A+B} - 2e^{-(A+B)}}{4} = \frac{e^{A+B} - e^{-(A+B)}}{2} = \sinh(A + B). \quad \checkmark$$

Proof of double angle formula $\cosh 2x = 2 \cosh^2 x - 1$:

$$\cosh 2x = \cosh(x + x) = \cosh^2 x + \sinh^2 x = \cosh^2 x + \cosh^2 x - 1 = 2 \cosh^2 x - 1. \quad \checkmark$$

Solving Hyperbolic Equations

Example 1 — Solve $\cosh x = 2$:

$$\frac{e^x + e^{-x}}{2} = 2 \Rightarrow e^x + e^{-x} = 4. \text{ Let } u = e^x > 0: u + \frac{1}{u} = 4 \Rightarrow u^2 - 4u + 1 = 0.$$

$$u = 2 \pm \sqrt{3}. \text{ Both positive: } x = \ln(2 + \sqrt{3}) \text{ or } x = \ln(2 - \sqrt{3}) = -\ln(2 + \sqrt{3}).$$

$$\text{So } x = \pm \ln(2 + \sqrt{3}) = \pm \operatorname{arccosh} 2. \quad \checkmark$$

Example 2 — Solve $3 \sinh x - \cosh x = 1$:

$$\text{Write in terms of exponentials: } \frac{3(e^x - e^{-x})}{2} - \frac{e^x + e^{-x}}{2} = 1.$$

$$\frac{3e^x - 3e^{-x} - e^x - e^{-x}}{2} = 1 \Rightarrow 2e^x - 4e^{-x} = 2 \Rightarrow e^x - 2e^{-x} = 1.$$

Let $u = e^x$: $u - \frac{2}{u} = 1 \Rightarrow u^2 - u - 2 = 0 \Rightarrow (u - 2)(u + 1) = 0$.
 $u = 2$ (since $u > 0$), so $x = \ln 2$.

Example 3 — Solve $\tanh x = \frac{1}{2}$:

$$\operatorname{arctanh}\left(\frac{1}{2}\right) = \frac{1}{2} \ln \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} = \frac{1}{2} \ln 3.$$

Inverse Hyperbolic Functions — Log Forms and Proofs

Function	Domain	Log form
$\operatorname{arcsinh} x$	$x \in \mathbb{R}$	$\ln(x + \sqrt{x^2 + 1})$
$\operatorname{arccosh} x$	$x \geq 1$	$\ln(x + \sqrt{x^2 - 1})$
$\operatorname{arctanh} x$	$ x < 1$	$\frac{1}{2} \ln \frac{1+x}{1-x}$

Proof that $\operatorname{arcsinh} x = \ln(x + \sqrt{x^2 + 1})$:

Let $y = \operatorname{arcsinh} x$, so $\sinh y = x$, i.e. $\frac{e^y - e^{-y}}{2} = x$.

$$e^{2y} - 2xe^y - 1 = 0. \text{ Quadratic in } e^y: e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}.$$

Since $e^y > 0$ and $\sqrt{x^2 + 1} > |x|$, take $+$: $e^y = x + \sqrt{x^2 + 1}$, so $y = \ln(x + \sqrt{x^2 + 1})$. ✓

Differentiation of Inverse Hyperbolic Functions

Function	Derivative
$\operatorname{arcsinh} x$	$\frac{1}{\sqrt{1+x^2}}$

$\operatorname{arccosh} x$	$\frac{1}{\sqrt{x^2 - 1}} \quad (x > 1)$
$\operatorname{arctanh} x$	$\frac{1}{1 - x^2} \quad (x < 1)$
$\operatorname{arcsinh} \frac{x}{a}$	$\frac{1}{\sqrt{a^2 + x^2}}$
$\operatorname{arccosh} \frac{x}{a}$	$\frac{1}{\sqrt{x^2 - a^2}} \quad (x > a)$

Proof that $\frac{d}{dx}(\operatorname{arcsinh} x) = \frac{1}{\sqrt{1 + x^2}}$:

Using log form: $\frac{d}{dx} \ln(x + \sqrt{x^2 + 1}) = \frac{1 + \frac{x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}} \cdot \checkmark$

Hyperbolic Substitution Integrals — Further Examples

Example 1: $\int \sqrt{x^2 + 4} \, dx$. Let $x = 2 \sinh u$, $dx = 2 \cosh u \, du$.

$$\sqrt{x^2 + 4} = \sqrt{4 \sinh^2 u + 4} = 2 \cosh u.$$

$$\begin{aligned} \int 2 \cosh u \cdot 2 \cosh u \, du &= 4 \int \cosh^2 u \, du = 4 \int \frac{1 + \cosh 2u}{2} \, du = 2u + \sinh 2u + C \\ &= 2u + 2 \sinh u \cosh u + C = 2 \operatorname{arcsinh} \frac{x}{2} + \frac{x \sqrt{x^2 + 4}}{2} + C \end{aligned}$$

Example 2: $\int_2^3 \frac{1}{\sqrt{x^2 - 1}} \, dx$. Let $x = \cosh u$, $dx = \sinh u \, du$.

$$\sqrt{x^2 - 1} = \sinh u. \text{ So } \int \frac{\sinh u}{\sinh u} \, du = u + C = \operatorname{arccosh} x + C.$$

$$[\operatorname{arccosh} x]_2^3 = \operatorname{arccosh} 3 - \operatorname{arccosh} 2 = \ln(3 + 2\sqrt{2}) - \ln(2 + \sqrt{3})$$

Osborn's Rule

Osborn's Rule: To convert a trigonometric identity to the corresponding hyperbolic identity, replace $\cos \theta$ with $\cosh \theta$, $\sin \theta$ with $i \sinh \theta$ (or equivalently: replace any product of two $\sin$ terms with minus the corresponding $\sinh$ product).

For example: $\cos^2 \theta + \sin^2 \theta = 1 \rightarrow \cosh^2 \theta - \sinh^2 \theta = 1$.

$\cos^2 \theta - \sin^2 \theta = \cos 2\theta \rightarrow \cosh^2 \theta + \sinh^2 \theta = \cosh 2\theta$.

Osborn's Rule caveat: It applies to basic identities, not to integrals or derivatives. Don't blindly apply it to results like $\int \sin^2 \theta d\theta$.

Graphs of Hyperbolic Functions

Key features:

- $\cosh x$: even function; minimum value 1 at $x = 0$; $\cosh x \rightarrow \infty$ as $x \rightarrow \pm\infty$. Graph is a catenary.
- $\sinh x$: odd function; passes through origin; strictly increasing; $\sinh x \approx x$ near 0.
- $\tanh x$: odd function; range $(-1, 1)$; horizontal asymptotes $y = \pm 1$; passes through origin with slope 1.
- $\operatorname{sech} x = 1/\cosh x$: even; max value 1 at $x = 0$; bell-shaped.
- $\operatorname{arcsinh} x$: odd; defined for all x ; looks like $\ln(2x)$ for large x .
- $\operatorname{arccosh} x$: defined for $x \geq 1$; starts at $(1, 0)$.
- $\operatorname{arctanh} x$: odd; defined for $|x| < 1$; vertical asymptotes at $x = \pm 1$.

Applications of Hyperbolic Functions — Mechanics

The Catenary: A flexible chain or cable hanging under gravity takes the shape $y = a \cosh \frac{x}{a}$. The arc length of the catenary from $x = 0$ to $x = L$ is:

$$s = a \sinh \frac{L}{a}$$

The tension at a point is $T = T_0 \cosh \frac{x}{a}$ where T_0 is the tension at the lowest point.

Example — Chain of length 20 m with lowest point 4 m below endpoints:

The chain hangs between poles distance $2d$ apart. With $y = a \cosh(x/a)$:

At $x = d$: $y = a + 4$ (4 m above lowest). Arc length $= 2a \sinh(d/a) = 20$, so $a \sinh(d/a) = 10$.

Also $a \cosh(d/a) = a + 4$. Dividing: $\tanh(d/a) = 10/(a + 4)$. Solve numerically.

Series Expansions of Hyperbolic Functions

Maclaurin series (all valid for all x):

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

Compare with: $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ and $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$.

The hyperbolic functions have all positive signs — consistent with $\cosh x = \cos(ix)$ and $\sinh x = -i \sin(ix)$.

Example — Approximate $\cosh(0.1)$:

$$\cosh(0.1) \approx 1 + \frac{0.01}{2} + \frac{0.0001}{24} = 1 + 0.005 + 0.0000042 \approx 1.0050042.$$

(Exact to 7 d.p.: 1.0050042.)

Connection: Trigonometric and Hyperbolic Functions

Using complex numbers ($i = \sqrt{-1}$):

$$\cos(ix) = \cosh x, \quad \sin(ix) = i \sinh x$$

$$\cosh(ix) = \cos x, \quad \sinh(ix) = i \sin x$$

This means: all hyperbolic identities follow from trig identities by the substitution $\theta \rightarrow ix$.

Example: Derive $\cosh(A + B) = \cosh A \cosh B + \sinh A \sinh B$ from $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$.

Set $\alpha = iA, \beta = iB$: $\cos(i(A + B)) = \cos(iA) \cos(iB) - \sin(iA) \sin(iB)$.

$\cosh(A + B) = \cosh A \cosh B - (i \sinh A)(i \sinh B) = \cosh A \cosh B + \sinh A \sinh B$.

✓

Complete Identities Summary

Identity	Value
$\cosh^2 x - \sinh^2 x$	1
$\operatorname{sech}^2 x + \tanh^2 x$	1
$\coth^2 x - \operatorname{csch}^2 x$	1
$\cosh 2x$	$\cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x$
$\sinh 2x$	$2 \sinh x \cosh x$
$\tanh 2x$	$\frac{2 \tanh x}{1 + \tanh^2 x}$
$\cosh(A + B)$	$\cosh A \cosh B + \sinh A \sinh B$
$\sinh(A + B)$	$\sinh A \cosh B + \cosh A \sinh B$

$$\tanh(A + B)$$

$$\frac{\tanh A + \tanh B}{1 + \tanh A \tanh B}$$

Integration by Parts Using Hyperbolic Functions

Example 1: $\int x \sinh x \, dx$

IBP: $u = x$, $dv = \sinh x \, dx \Rightarrow v = \cosh x$.

$$= x \cosh x - \int \cosh x \, dx = x \cosh x - \sinh x + C$$

Example 2: $\int x \cosh x \, dx$

IBP: $u = x$, $dv = \cosh x \, dx \Rightarrow v = \sinh x$.

$$= x \sinh x - \int \sinh x \, dx = x \sinh x - \cosh x + C$$

Example 3: $\int \sinh^2 x \, dx$

Use $\sinh^2 x = \frac{\cosh 2x - 1}{2}$:

$$\int \sinh^2 x \, dx = \frac{\sinh 2x}{4} - \frac{x}{2} + C = \frac{\sinh x \cosh x}{2} - \frac{x}{2} + C$$

Example 4: $\int \cosh^2 x \, dx$

Use $\cosh^2 x = \frac{\cosh 2x + 1}{2}$:

$$\int \cosh^2 x \, dx = \frac{\sinh 2x}{4} + \frac{x}{2} + C = \frac{\sinh x \cosh x}{2} + \frac{x}{2} + C$$

Further Worked Differentiation Examples

Example 1 — Chain rule: $\frac{d}{dx} \cosh(3x^2 + 1)$

$$= \sinh(3x^2 + 1) \cdot 6x = 6x \sinh(3x^2 + 1)$$

Example 2 — Product rule: $\frac{d}{dx}[x^2 \sinh x]$
 $= 2x \sinh x + x^2 \cosh x$

Example 3 — Implicit differentiation: Given $\cosh y = x$, find $\frac{dy}{dx}$.

$$\sinh y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sinh y}.$$

Since $\cosh y = x$: $\sinh y = \sqrt{x^2 - 1}$ (for $y \geq 0$).

So $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$ — confirming the derivative of $\operatorname{arccosh} x$.

Hyperbolic Functions — AQA Exam Technique

- **Equations:** If given $\cosh x = k$ with $k > 1$, write $\pm \operatorname{arccosh} k = \pm \ln(k + \sqrt{k^2 - 1})$. Both signs are valid unless domain restricted.
- **Integration:** Recognise $\frac{1}{\sqrt{x^2 \pm a^2}}$ and use $\operatorname{arcsinh}/\operatorname{arccosh}$. For $\frac{1}{a^2 - x^2}$, use $\operatorname{arctanh}$ (when $|x| < a$).
- **Completing the square:** If the denominator is $\sqrt{x^2 + 4x + 8} = \sqrt{(x + 2)^2 + 4}$, substitute $u = x + 2$ first.
- **Identity derivation:** Always go back to definitions e^x if unsure — expand both sides from scratch.
- **Log form answers:** AQA often accepts both $\operatorname{arcsinh}/\operatorname{arccosh}$ notation and the logarithmic form. Show they are equivalent for full marks.
- **Osborn's Rule:** Useful for guessing identities, but always verify from definitions for a proof question.

Completing the Square and Hyperbolic Integration

Example 1: $\int \frac{1}{\sqrt{x^2 + 6x + 10}} dx$

$x^2 + 6x + 10 = (x + 3)^2 + 1$. Let $u = x + 3$:

$$\int \frac{1}{\sqrt{u^2 + 1}} du = \operatorname{arcsinh} u + C = \ln(x + 3 + \sqrt{x^2 + 6x + 10}) + C$$

Example 2: $\int \frac{1}{\sqrt{x^2 - 4x + 3}} dx$

$x^2 - 4x + 3 = (x - 2)^2 - 1$. Let $u = x - 2$:

$$\int \frac{1}{\sqrt{u^2 - 1}} du = \operatorname{arccosh} u + C = \ln(x - 2 + \sqrt{x^2 - 4x + 3}) + C \quad (x > 3)$$

Example 3: $\int \frac{1}{3 - x^2} dx \quad (|x| < \sqrt{3})$

Write as $\frac{1}{(\sqrt{3})^2 - x^2}$. Use $\operatorname{arctanh}$ formula:

$$= \frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3} + x}{\sqrt{3} - x} + C = \frac{1}{\sqrt{3}} \operatorname{arctanh} \frac{x}{\sqrt{3}} + C$$

Hyperbolic Functions — Numerical Values

Useful values to remember or compute quickly:

- $\cosh 0 = 1$, $\sinh 0 = 0$, $\tanh 0 = 0$
- $\cosh 1 \approx 1.543$, $\sinh 1 \approx 1.175$, $\tanh 1 \approx 0.762$
- $\operatorname{arcsinh} 1 = \ln(1 + \sqrt{2}) \approx 0.881$
- $\operatorname{arccosh} 2 = \ln(2 + \sqrt{3}) \approx 1.317$
- $\operatorname{arctanh} \frac{1}{2} = \frac{1}{2} \ln 3 \approx 0.549$
- As $x \rightarrow \infty$: $\cosh x \approx \sinh x \approx \frac{e^x}{2}$, $\tanh x \rightarrow 1$.
- The parametric equations $x = \cosh t$, $y = \sinh t$ satisfy $x^2 - y^2 = 1$, tracing the right branch of the unit hyperbola — this is the geometric origin of the name "hyperbolic functions".
- Compare with circular functions: $x = \cos t$, $y = \sin t$ satisfies $x^2 + y^2 = 1$ (unit circle). The parameter t in the hyperbolic case represents twice the area swept out, not an angle.

- For AQA, you must know: all basic derivatives and integrals of $\sinh$, $\cosh$, $\tanh$ and their inverses; how to prove identities from e^x definitions; and how to solve equations using the quadratic-in- e^x technique.
- The identity $\cosh^2 x - \sinh^2 x = 1$ is the hyperbolic analogue of $\cos^2 \theta + \sin^2 \theta = 1$. Use it whenever you need to simplify $\cosh^2$ or $\sinh^2$ — analogous to how $\sin^2 + \cos^2 = 1$ is used in trigonometry.